



Weakly Supervised Breast Ultrasound Image Segmentation Based on Image Selection

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Abstract

Automatic image segmentation in Breast Ultrasound (BUS) imaging is vital to BUS computer-aided diagnostic systems. Fully supervised learning methods achieve high accuracy but require hard-to-obtain pixel-level annotations. As an alternative, weakly supervised learning reduces this extensive annotation dependency and offers a more practical approach for training models.

In this research, we proposed a novel framework by selection high quality candidate for pseudo label generation. Our framework is evaluated on a public BUS image dataset and achieves an Intersection over Union (IoU) score that is 82.9% of what is attained by fully supervised methods.

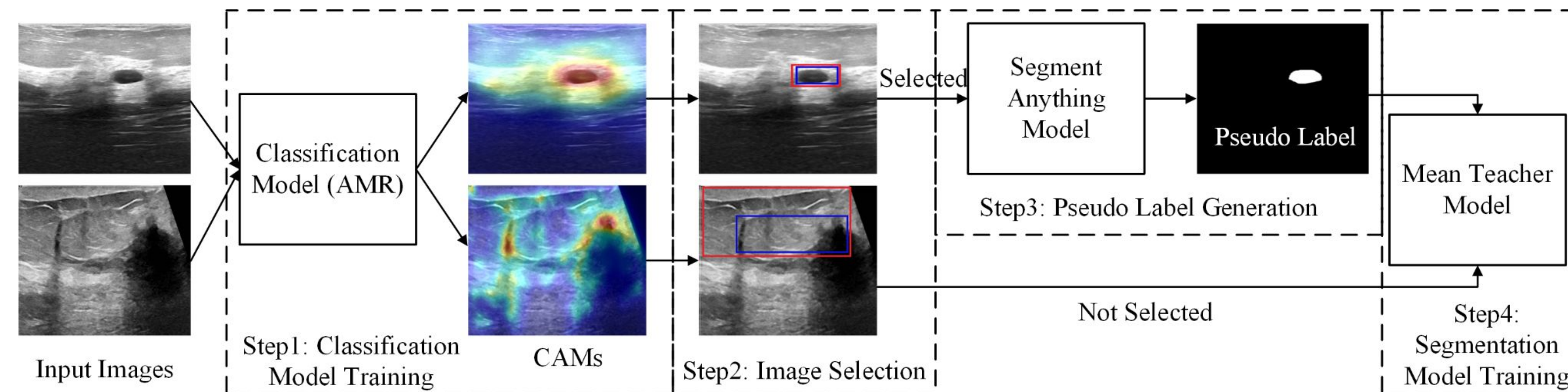
Methods and Materials

The proposed method comprises four key steps:

- 1) Training a classification model using the entire dataset, denoted as V , and generating class activation maps (CAMs) for all images in V ;
- 2) Employing the proposed selection method (Algorithm 1) to identify potential candidate images that are likely to yield high-quality pseudo labels and placing these selected images into a subset L , while assigning the remaining images to subset U , where $V = L + U$;
- 3) Producing pseudo-segmentation labels for images within L using "Segment Anything";
- 4) Training a U-Net segmentation model using the Mean Teacher method, utilizing both subsets L and U .

A detailed illustration of our proposed approach is depicted in Fig. 1.

Methods and Materials



Algorithm 1 Pseudo Label Generation Candidate Selection

Input: Normalized $CAM \in \mathbb{R}^{W \times H}$ in the tumor class, percentage $\alpha\%$.
 1: Calculate histogram $H(i)$ for $i = 0$ to 255.
 2: Calculate cumulative sum $C(i)$: $C(i) = \sum_{j=1}^{255} H(j)$, for $i = 255$ to 0.
 3: Calculate total number of pixels P_{total} in CAM .
 4: Find the minimum intensity T : $argmin_T (C(T) \leq \alpha\% \times P_{total})$.
Output: Intensity T .

Fig 1. An illustration of the proposed method

Experiments and Results

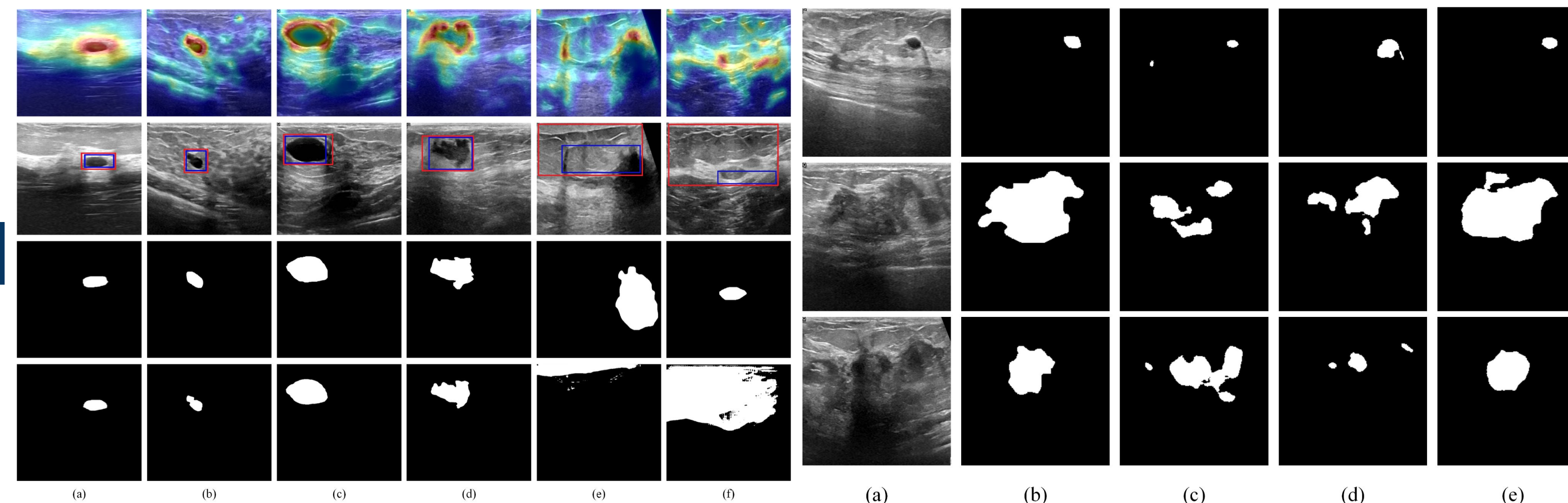


Fig 2. Three distinct categories of images, their corresponding CAMs, real labels, and pseudo labels: The first row displays the CAMs produced by the Activation Modulation and Recalibration (AMR) model. The second row shows bounding boxes created using two thresholds: red boxes result from a lower threshold, while blue boxes derive from a higher threshold. The third row illustrates the real segmentation labels. Finally, the fourth row depicts pseudo-segmentation labels generated by the 'Segment Anything' model.

Fig 3. Semantic segmentation results:

(a) Original BUS images, (b) Real segmentation ground truths, (c) Pseudo-Subset, (d) Pseudo-Whole, (e) Pseudo-MeanTeacher

Methods	Tumor IoU	Background IoU	mIoU
GT-Whole	67.9%	94.8%	81.3%
GT-Subset	56.3%	93.9%	75.1%
GT-MeanTeacher	64.0%	94.1%	79.0%
Pseudo-Whole	48.5%	91.7%	70.1%
Pseudo-Subset	35.2%	91.1%	63.1%
Pseudo-MeanTeacher	56.3%	91.3%	73.8%

Table 1. Ablation study results

Methods	Tumor IoU	Background IoU	mIoU
VGG16-CAM [14]	24.1%	89.1%	56.6%
ResNet50-CAM [14]	37.2%	91.4%	64.3%
OAA-CAM [15]	30.4%	85.3%	57.8%
Swin MIL [12]	25.4%	89.3%	57.6%
ReCAM [22]	31.7%	88.1%	59.9%
AMR [16]	35.8%	87.5%	61.7%
Pseudo-MeanTeacher (Ours)	56.3%	91.3%	73.8%

Table 2. Quantitative comparison to other methods

Conclusion

In response to the high costs of medical image annotation, we introduce a novel weakly supervised segmentation method for BUS images that minimizes annotation needs while maintaining high performance. This method selects high-quality candidates for pseudo-label generation and employs the Mean Teacher approach for training on pseudo-labeled and unlabeled images. Our method reduces annotation costs and labor, crucial for medical imaging's precision and reliability. Tested on a public BUS dataset, it surpasses six recent weakly supervised methods, achieving state-of-the-art results.

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